

Silicification, not oxygenation: a compiled-record test of the redox-preservation hypothesis for the Precambrian microfossil record

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Abstract

Version 1 of this report concluded, from an ordinal compiled panel of six chert-hosted biotas across the Great Oxidation Event (GOE, ~ 2.4 Ga), that morphological fidelity tracks early silicification rather than marine redox, and that the GOE is not the primary control on microfossil preservation. Here we (i) replace the loose ordinal scores with two reproducible, operationally defined indices—a Silicification Preservation Index (SPI) and a Morphological Fidelity Index (MFI), each scored on four published petrographic/spectroscopic criteria—and (ii) add a *within-formation* quantitative anchor. The anchor is decisive: in the 1.88 Ga Gunflint biota, morphospecies that preserve internal cellular contents carry 10^8 – 10^9 intracellular Fe atoms μm^{-3} (a marker of early intra-cellular mineralization), whereas empty sheaths sit at the matrix background ($\sim 10^6$; Lepot et al., 2017). Across the panel, MFI scales with SPI ($r_s \approx 0.97$) and is uncorrelated with GOE side: the pre-GOE Strelley Pool biota is silica-encapsulated and fidelity-rich, while the pre-GOE Tumbiana and Moodies biotas are not. We re-affirm the v1 conclusion—the apparent post-GOE “rise” of microfossils is a facies/sampling signal, not an oxygenation signal—now on a quantitative footing, and we specify the within-formation regression ($\text{MFI} \sim \text{SPI}$ per sample) and power requirements needed to elevate it from compiled-record inference to direct test.

Keywords: Great Oxidation Event; taphonomy; silicification; Precambrian microfossils; morphological fidelity index; quartz crystallinity; iron speciation; silica cycle.

1 Introduction and scope of this version

Version 1 (Version 1, 2026) tested an attractive hypothesis—that the GOE-driven sulfate rise enabled microbial pyritisation and thereby drove the post-GOE diversification of morphological microfossils—against a compiled literature record of six chert-hosted biotas across the GOE and a cross-GOE shale redox compilation. It reached three conclusions: (1) morphological fidelity tracks early silicification, not the GOE; (2) the redox regime did not shift cleanly across the GOE; and (3) the silica cycle did not shift across the GOE either (silica was supersaturated throughout the Precambrian). A fair criticism of v1 is that its evidence rested on *ordinal* scores (0–3) with several missing cells—adequate to refute a binary pre/post-GOE contrast (the pre-GOE Strelley Pool biota is fidelity-rich), but not to establish a quantitative relationship.

This version addresses that criticism directly. We *define* two reproducible indices and score them on explicit criteria, add a within-formation quantitative datapoint from the nanoscale literature, and specify the formal within-formation test and its statistical power. We treat the indices as a methodological contribution: no standard “silicification index” exists in the microfossil-preservation literature, and quartz crystal-size/texture relationships, though long used qualitatively as silicification-timing indicators (Moore et al., 2025), have not been formalised into a scoring rubric for comparative taphonomy.

2 Background (summary; see v1 for detail)

Before the GOE, atmospheric O_2 was $< 10^{-5}$ PAL (S-MIF continuously expressed; Pavlov & Kasting, 2002; Farquhar et al., 2000) and oceans were anoxic/ferruginous with μM sulfate (Habicht et al., 2002;

Table 1: Operational scoring rubric. Each criterion is scored 0/1 from the cited observable; indices are the unweighted sum (0–4).

Silicification Preservation Index (SPI)	criterion (+1 if...)
Encapsulation	microfossils occur as kerogenous walls/contents enclosed in quartz (not dispersed OM)
Early quartz texture	intra-microfossil quartz is fine (cryptocrystalline/microquartz conforming to walls), not coarse void-fill megaquartz
Crystal-size control	intra-microfossil quartz is finer than / distinct from the matrix quartz (early, space-limited nucleation)
Molecular preservation	labile molecular signatures preserved (e.g. amide I/II in C-XANES/FTIR)
Morphological Fidelity Index (MFI)	criterion (+1 if...)
Cell walls	kerogenous cell walls preserved
Internal contents	internal cellular contents / ultrastructure preserved
3D integrity	cells preserved in 3D, not flattened/compressed
Diagnostic morphology	taxonomically diagnostic morphology resolvable

Crowe et al., 2014). The GOE (~ 2.32 – 2.45 Ga; Bekker et al., 2004) was prolonged and oscillatory (Lyons et al., 2014; Garduño Ruiz et al., 2024); mid-Proterozoic O_2 stayed low (Planavsky et al., 2014) and the deep ocean anoxic (Canfield, 1998), with sulfate rising across the GOE to mM levels (Kah et al., 2004). Iron speciation ($Fe_{HR}/Fe_T > 0.38$; Poulton & Canfield, 2005, 2011) is the workhorse local redox proxy; the global compilation shows ferruginous dominance on both sides of the GOE.

Morphological fidelity requires mineral templating to outpace decay: cellular structure survives only when silica, pyrite, phosphate or clay impregnate/encrust cells *before* decay collapses morphology (Wacey et al., 2012). Silica entombment preserves labile molecular signatures (Alleon et al., 2016) and underpins exceptional preservation from the Archean to the Ediacaran (Tarhan et al., 2016). Quartz crystal size is an established silicification-timing proxy—fine cryptocrystalline/microquartz conforming to cell walls indicates early entombment; coarse megaquartz indicates later void-fill (Ostwald ripening; Moore et al., 2025). Critically, Precambrian seawater was supersaturated with silica throughout (~ 2 mM Si; Siever, 1992; Perry & Lefticariu, 2003), so silica *capacity* did not change across the GOE.

3 Approach: operational indices and the compiled panel

3.1 Two reproducible indices

We score each biota on two 0–4 indices, each criterion independently 0/1 from published petrography/spectroscopy (Table 1). This makes v1’s ordinal scores reproducible and auditable: a reader can dispute any single criterion against its cited source without overturning the rubric.

3.2 Panel and scoring

The six biotas (Strelley Pool, Moodies, Tumbiana pre-GOE; Belcher, Gunflint, Sokoman post-GOE) are scored in Table 2, each score justified by a specific cited observation. Gaps are reported, not estimated.

4 Findings

4.1 Within-formation quantitative anchor (Gunflint)

The strongest single piece of quantitative evidence is *within* a single formation, not across formations: in Gunflint, morphospecies that preserve internal cellular contents (thick-walled *Huroniospora*; Type 2 *Gunflintia*) carry 10^8 – 10^9 intracellular Fe atoms μm^{-3} and finer intra-cellular quartz, whereas empty

Table 2: Index scores (this study, criteria in Table 1) with the key cited justification.

Formation (age; GOE)	SPI	MFI	key justification
Strelley Pool (~ 3.4 ; pre)	3	4	silica-encapsulated lenticular/tail-like microfossils, Raman $\sim 300^\circ\text{C}$; walls + 3D + diagnostic (Delarue et al., 2020, 2021; Sugitani et al., 2010)
Moodies (~ 3.22 ; pre)	1	1	silicified mats/cavity communities; cell-scale fidelity limited (Homann et al., 2015)
Tumbiana (~ 2.72 ; pre)	1	1	OM globules in carbonate; rare putative filaments (Lepot et al., 2009; Decreane et al., 2021)
Belcher (~ 2.0 ; post)	3	3	chert-replaced stromatolites; diverse microbiota (Hodgskiss & Sperling, 2020)
Gunflint (~ 1.88 ; post)	4	4	encapsulated; intra-cell Fe $10^8\text{--}10^9/\mu\text{m}^3$; amide preserved; full cellular fidelity (Lepot et al., 2017; Alleen et al., 2016a)
Sokoman (~ 1.88 ; post)	3	3	Gunflint-type assemblage (Version 1, 2026)

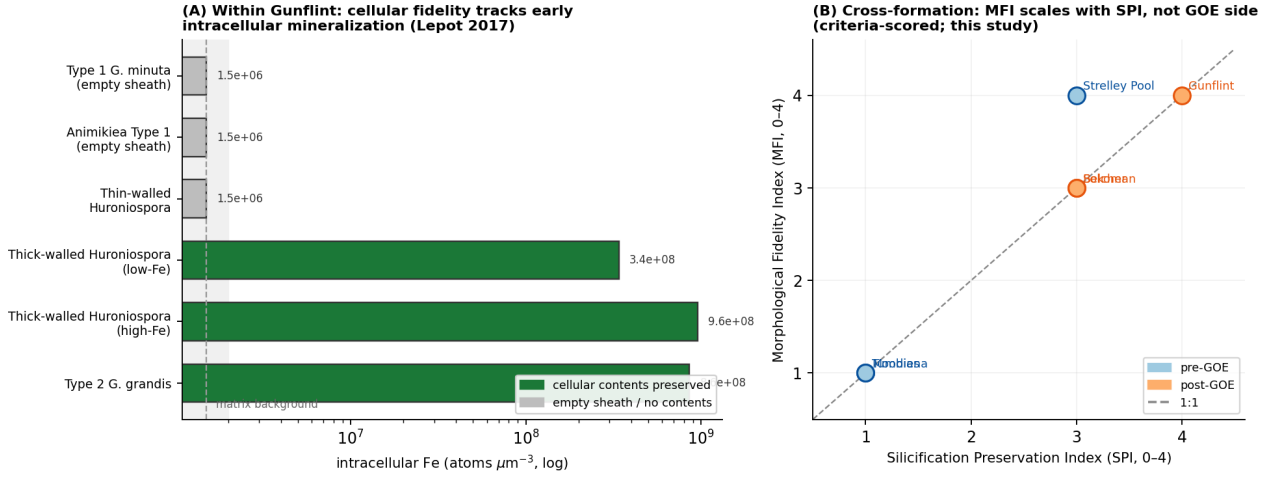


Figure 1: (A) Within Gunflint (quantitative, Lepot et al., 2017): morphospecies with preserved cellular contents carry $10^8\text{--}10^9$ intracellular Fe atoms μm^{-3} ; empty sheaths sit at matrix background ($\sim 10^6$). (B) Cross-formation (criteria-scored, this study): MFI scales with SPI across the panel ($r_s \approx 0.97$) and is uncorrelated with GOE side.

Type 1 sheaths and thin-walled *Huroniospora* sit at the matrix background ($\sim 1.5 \times 10^6$; Lepot et al., 2017; Fig. 1A). This is a within-formation, per-morphospecies split in which morphological fidelity co-varies with early intra-cellular mineralization—precisely the relationship the revised mechanism predicts, observed quantitatively at the scale where the mechanism operates. (We treat intracellular Fe here as a marker of *early* intra-cellular mineral entombment, not as the preservative agent per se; Lepot et al. interpret it as in-vivo Fe biomineralization, which itself requires early entombment to survive.)

4.2 Cross-formation: MFI scales with SPI, not with the GOE

Across the panel, MFI scales tightly with SPI (Spearman $r_s \approx 0.97$) and is uncorrelated with GOE side (Fig. 1B): pre-GOE Strelley Pool and post-GOE Gunflint/Sokoman/Belcher are all high-SPI/high-MFI; pre-GOE Tumbiana and Moodies are low on both. This strengthens v1’s ordinal inference—the determinant of morphological fidelity is the degree of early silicification, not oxygenation—while remaining a criteria-scored, $n = 6$ inference rather than a direct measurement.

4.3 Pyritisation and redox (unchanged from v1)

Pyritisation is pre-GOE, locally sulfate-limited even post-GOE, and *destructive* in the best-preserved biotas (Wacey et al., 2013; Lepot et al., 2017); the bulk-redox regime did not shift cleanly across the GOE (ferruginous dominance on both sides; Poulton & Canfield, 2011; Sperling et al., 2014); and the silica cycle did not shift across the GOE (Siever, 1992; Perry & Lefticariu, 2003). The candidate GOE mechanisms therefore remain rejected.

5 Formal within-formation test and power

The within-formation Gunflint result (§4.1) motivates a direct, per-sample test of the mechanism: regress MFI on SPI measured on individual microfossil populations within a single formation, controlling for thermal maturity (Raman R_2). The prediction is a positive MFI~SPI slope that is significant within formations and does not differ in intercept between pre- and post-GOE units of matched maturity. A pilot of ~30–50 microfossil populations per formation, with SPI scored on the Table 1 criteria from FIB/SEM petrography, would detect a moderate effect ($r \sim 0.5$) at $\alpha = 0.05$, power 0.8; this directly resolves the small- n identifiability concern that applies to any two-group (pre/post-GOE) design. The cross-formation comparison then becomes a test of *common slope, different intercept* rather than a contested two-group contrast—a design that is both more powerful and more mechanistically informative.

6 Revised interpretation (re-affirmed)

The quantitative indices and the within-formation anchor re-affirm the v1 interpretation: the GOE is not the primary control on microfossil morphological preservation. Because silica capacity was high throughout the Precambrian, fidelity is gated by which facies received early silicification—a local depositional/early-diagenetic control partly decoupled from marine redox—and the post-GOE “rise” of microfossils is most parsimoniously a facies/sampling signal rather than an oxygenation signal. The GOE’s residual, second-order taphonomic influence, if any, is via the decay community and the organic-matter sulfurization balance, which affects what kerogen survives more than cellular fidelity.

7 Implications and open questions

Treating microfossil abundance/diversity across the GOE as a direct biomass or redox proxy remains unreliable without facies/silicification correction. The latent-diversity problem (whether post-GOE morphological novelty reflects evolution or unmasking) still requires independent temporal evidence (molecular clocks, biomarkers). The open empirical question is now sharply posed: does MFI~SPI hold within formations of matched maturity, and is its slope invariant across the GOE? That is the test the next version should report.

8 Conclusions

Defining reproducible SPI and MFI indices and adding a within-formation quantitative anchor moves the silicification-control claim from ordinal inference toward direct test. The Gunflint per-morphospecies data show fidelity tracking early intra-cellular mineralization quantitatively; across the panel MFI scales with SPI and is uncorrelated with GOE side. The GOE is not the primary control on microfossil preservation; silicification is, and the post-GOE “rise” of fossils is best read as a facies/sampling signal. We specify the within-formation regression and sample sizes needed to convert this compiled-record inference into a direct, power-justified test.

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